

Mel-Frequency Cepstral Coefficients and Neural Networks for Indonesian Traditional Music Recognition

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Abstract— This study addresses the challenge of automated music recognition in the context of culturally specific soundscapes, focusing on Indigenous Indonesian Music recognition. We propose a hybrid deep learning model, Convolutional Neural Network (CNN) combined with Long Short-Term Memory (LSTM), to effectively classify two distinct Indonesian genres: Dangdut and Keroncong. The methodology involves extracting Mel-Frequency Cepstral Coefficients (MFCCs) along with their delta and delta-delta coefficients to capture both spectral and temporal characteristics of the audio signals. The CNN component is designed to learn robust local features from the MFCC spectrograms, while the subsequent LSTM layer processes these spatial features over time, modeling the complex rhythmic and melodic sequences inherent in the music. Utilizing a curated dataset of 1,687 recordings, the proposed CNN-LSTM hybrid model achieved a consistent classification accuracy of 70.3 percent on the test set. Further evaluation yielded an F1-Score of 0.69 and a Matthews Correlation Coefficient (MCC) of 0.41. This result establishes a viable baseline for this challenging indigenous cultural music information retrieval (MIR) task. This research validates the utility of a combined cepstral-temporal approach for complex music classification and highlights the need for dedicated, high-quality, and larger datasets in cultural AI applications.

Keywords— MFCC, CNN-LSTM Hybrid, Audio Classification, Cultural Music Information Retrieval (MIR), Temporal Feature Modeling

I. INTRODUCTION

Indonesia possesses a rich diversity of traditional music, characterized by unique rhythms, tonal structures, and instrumental arrangements that reflect the nation's cultural identity. Among these, Dangdut and Keroncong are two of the most prominent genres, each exhibiting distinct acoustic signatures and performance styles [1]. The preservation and automatic recognition of such complex, culturally-specific music have become increasingly important in the digital era, yet they present significant challenges for standard Music Information Retrieval (MIR) systems due to rhythmic irregularities and limited annotated data [2].

Automatic music recognition has been widely explored in the general field of MIR, enabling genre classification and content analysis [3]. Traditional approaches often rely on handcrafted features (such as MFCC) combined with shallow learning models, which may not effectively capture the complex temporal and spectral dynamics inherent in traditional music. Recent years have seen the dominance of Deep Learning in audio classification, leveraging its ability to learn hierarchical representations directly from features like Mel-Frequency Cepstral Coefficients (MFCCs) [4]. While powerful, standard approaches like purely CNN-based models often excel at spatial feature extraction but frequently neglect long-range temporal dependencies, a critical aspect of identifying genres defined by rhythm and melody sequences.

Despite the extensive progress in general MIR, research focusing on Indigenous Indonesian Music Recognition remains critically limited [5]. Most prior works either target global Western genres (like Rock or Pop) or utilize simpler machine learning models for local music, failing to address the intricate complexity of genres like Dangdut and Keroncong using state-of-the-art architectures. Crucially, there is an explicit scientific gap in developing systems that can efficiently model both spectral characteristics (which CNNs handle well) and sequential temporal relationships (which LSTMs handle well) simultaneously within this unique cultural domain. This lack of focused, advanced modeling for Indonesian music prevents the establishment of a robust, high-quality technical baseline necessary for future cultural AI development.

This study aims to fill the identified gap by proposing an automatic recognition model for Dangdut and Keroncong utilizing an MFCC-based hybrid CNN-LSTM architecture. We specifically choose the CNN-LSTM hybrid over simpler models (CNN-only) or more complex, computationally intensive models (such as Audio Transformers), because the CNN-LSTM provides an optimal balance: the CNN component efficiently extracts local, frequency-invariant patterns from the MFCC spectrograms, while the subsequent LSTM layer is utilized to model the long-term temporal

dependencies across these feature sequences [6]. This dual-capability model is essential for classifying genres where rhythmic and melodic continuity is paramount. The specific objectives of this research are: 1) To establish a technical baseline for Dangdut and Keroncong classification using the proposed hybrid model. 2) To evaluate the model's performance using standard metrics (Accuracy, F1-Score, and MCC)

The remainder of this paper is structured as follows. Section 2 discusses related work and compares previous approaches. Section 3 details the methodology, including data preparation, feature extraction, and the mathematical formulation of the hybrid CNN-LSTM architecture. Section 4 presents the experimental setup and results, including a comparative analysis against single-model baselines. Section 5 discusses the findings and error analysis, and Section 6 concludes the paper and outlines directions for future research..

II. RELATED WORK

Automatic music recognition (AMR) and genre classification have been active research areas within Music Information Retrieval (MIR). The trajectory of research in this field has evolved from classical feature engineering to deep learning-based representation learning.

A. Feature Engineering and Classical Machine Learning

Early studies in AMR relied primarily on classical machine learning algorithms such as K-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and Random Forests, utilizing handcrafted features including spectral centroid, chroma, and Mel-Frequency Cepstral Coefficients (MFCCs) [7][8]. While these methods achieved reasonable performance for small-scale datasets with simple genre boundaries (e.g., distinguishing classical from rock), they often failed to generalize effectively to highly complex and rhythmically diverse music, which includes many traditional and indigenous genres. The use of MFCCs, however, remains the foundational step due to their capability to approximate human auditory perception and effectively represent timbral characteristics [9]. For instance, the pioneering work by Tzanetakis and Cook [10] established a benchmark by proposing a classification framework using a combination of timbral, rhythmic, and pitch-based features. However, classical models struggle to capture the non-linear long-term dependencies crucial for identifying subtle differences between closely related folk genres.

B. Deep Learning Architectures in Audio Classification

Deep learning has advanced AMR through models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) [11]. CNNs excel at extracting local, translational-invariant spectral patterns from spectrograms, making them powerful feature compressors. Conversely, Long Short-Term Memory (LSTM) networks are specialized for modeling sequential temporal dependencies, which is critical for capturing rhythmic and melodic continuity in music data [12].

Recent literature emphasizes hybrid architectures that combine CNNs and LSTMs, which have demonstrated state-of-the-art results in tasks like speech recognition and instrument classification where both spectral content and temporal context are essential [13]. This hybrid approach is

generally superior to pure CNN-based models, such as those applied in some Chinese folk music studies [14], which often overlook the necessary sequential feature modeling.

C. Research Gap and Positioning

Despite the advancements, the application of deep learning to Indigenous Indonesian Music remains under-explored, particularly regarding temporal modeling. Preliminary work by Tomo et al. [15] used MFCCs with SVM for gamelan instrument recognition, setting an initial local benchmark but utilizing a shallow learning approach.

Our work addresses the critical scientific gap concerning the lack of dedicated deep learning baselines with temporal modeling for Dangdut and Keroncong. While other studies utilize related models like CRNNs for instrument classification [16], the specific characteristics of these Indonesian genres; especially their complex, sequential rhythmic patterns; necessitate the explicit use of the hybrid CNN-LSTM. This combination allows for a robust methodology to model both spectral characteristics (via CNN) and sequential temporal dependencies (via LSTM) [17]. The contribution of this study is the establishment of a technical baseline for Dangdut and Keroncong classification using the proposed hybrid architecture, thus advancing cultural MIR research in the national context.

III. METHOD

The methodological framework combines digital signal processing and a hybrid deep learning architecture for the automatic classification of Indonesian traditional music. The pipeline ensures robust feature extraction, effective temporal dependency learning, and rigorous performance validation [18][19]. The overall research flow is summarized in Figure 1, detailing the systematic process from data acquisition to evaluation.

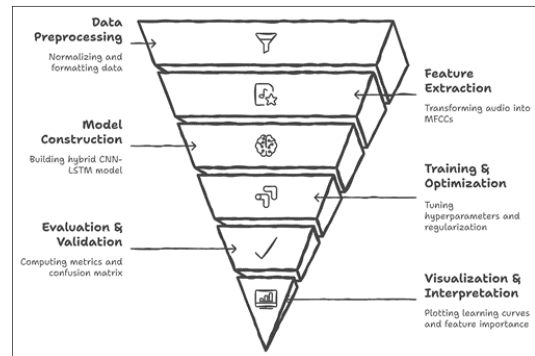


Fig 1. Overall Research Framework of the Hybrid CNN-LSTM System.

The funnel diagram illustrates the systematic flow, beginning with Data Preprocessing, Feature Extraction (MFCC), Model Construction, Training & Optimization, and concluding with Evaluation & Interpretation.

A. Dataset Description and Data Augmentation

The curated dataset consists of 1,687 recordings of Indigenous Indonesian Music, categorized into two genres: 444 Dangdut and 1,243 Keroncong.

- **Preprocessing and Standardization:** All raw audio files were standardized to mono-channel, resampled to 22,050 Hz, and segmented into 30-second clips (with padding or truncation as necessary) to ensure uniform

input length. Signals were normalized using min-max scaling and stored in 16-bit WAV format [20].

- **Stratified Split:** To maintain class distribution parity and mitigate the existing imbalance (Keroncong:Dangdut ratio), a Stratified Sampling approach was utilized to split the dataset into an 80% Training Set and a 20% Test Set [21].
- **Data Augmentation:** To address the small dataset size and mitigate overfitting, Data Augmentation was applied exclusively to the training set. Techniques employed include Pitch Shifting and Time Stretching to enhance the model's robustness and generalization capability [22].

B. Feature Extraction: MFCC and Mathematical Formulation

Feature extraction transforms the raw audio signal $x(t)$ into perceptually relevant representations. This study uses the Mel-Frequency Cepstral Coefficients (MFCCs) due to their effectiveness in approximating human auditory perception of timbre [23]. The detailed steps of the audio signal conversion into features is visualized in Figure 2.

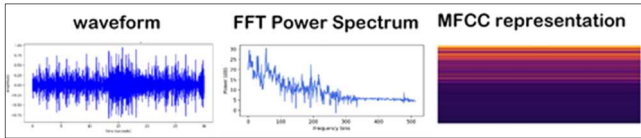


Fig 2. Audio Preprocessing and Feature Generation Pipeline.

Visualization of the conversion process from raw audio (Waveform) into the frequency domain (FFT Power Spectrum) and finally to the dense feature representation (MFCC Spectrogram) used as input for the deep learning model [24]. The MFCC computation involves the following key steps, including the Discrete Cosine Transform (DCT) to yield the MFCC vector ζ_1 :

$$\zeta_1 = \sum_{k=1}^K (\log E_k) \cos \left[i \left(k - \frac{1}{2} \right) \frac{\pi}{K} \right] \text{ for } i = 1, 2, \dots \quad (1)$$

In this study, thirteen MFCC coefficients ($D=13$) are extracted per frame, augmented with their first-order (Δ) and second-order ($\Delta\Delta$) derivatives, resulting in a 39-dimensional feature vector per frame. The effectiveness of these features in separating the genres is confirmed visually in Figure 3 and Figure 4. Illustrates clear differences in the spectral envelopes captured by the MFCC features.

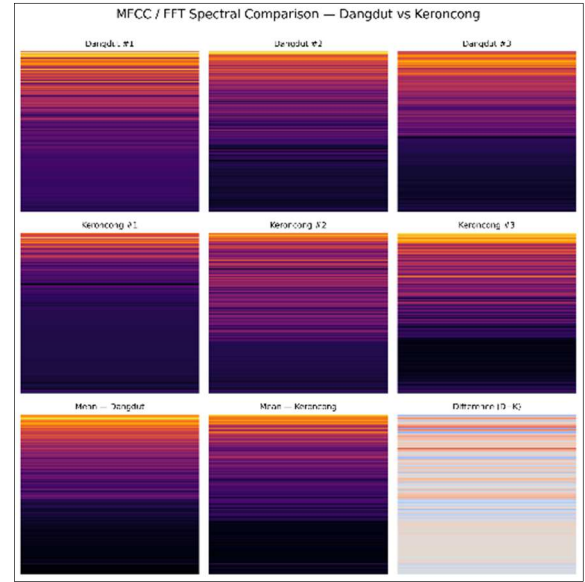


Fig 3. Mean MFCC Distribution Comparison (Dangdut and Keroncong).

Highlights the stronger low- to mid-frequency energy concentration in Dangdut music compared to the more balanced distribution in Keroncong recordings

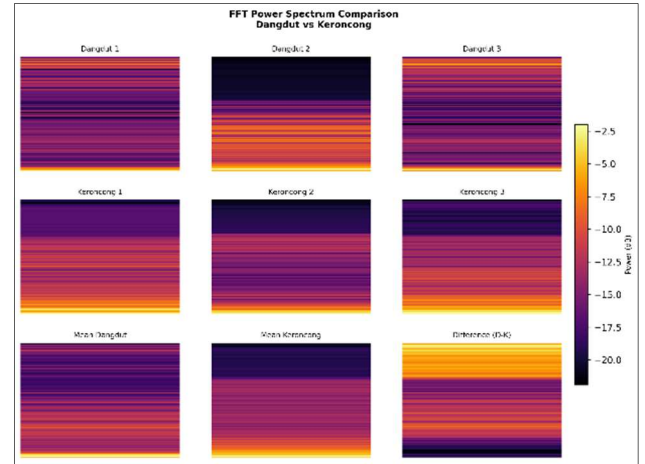


Fig 4. FFT Power Spectrum Comparison (Average).

C. Model Architecture: Hybrid CNN-LSTM and Temporal Formulation

To leverage both spectral and temporal characteristics, this study implements a hybrid Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) architecture [25], illustrated in Figure 5.

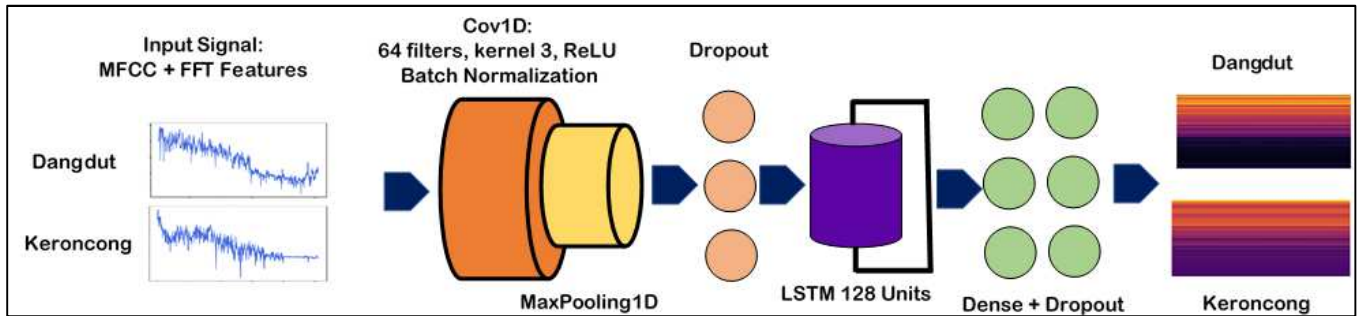


Fig 5 Architecture of the proposed CNN-LSTM model.

The LSTM layer is crucial for modeling long-term dependencies in the sequences outputted by the CNN block.

The LSTM unit at time step t is governed by three primary gates “the Forget Gate” f_t , “the Input Gate” I_t , and “the Output Gate” O_t , defined as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t]) + b_f \quad (2)$$

$$C_t = f_t * C_{t-1} + I_t * \tilde{C}_t \quad (3)$$

The detailed layer structure, output shapes, and parameter counts are summarized in Table 1.

TABLE I. HYBRID CNN-LSTM MODEL LAYER AND PARAMETER DETAILS

Layer	Units	Kernel	Activation	Output	Parameters
Input Layer	-	-	-	(Time Steps, 39)	0
Conv2D (Block 1)	64	3x3	ReLU	(Time Steps, Freq, 64)	5632
MaxPooling2D	-	2x2	-	(Time Steps/2, Freq/2, 64)	0
Dropout	0.25	-	-	(Time Steps/2, Freq/2, 64)	0
Conv2D (Block 2)	128	3x3	ReLU	(Time Steps/2, Freq/2, 128)	73,856
MaxPooling2D	-	2x2	-	(Time Steps/4, Freq/4, 128)	0
Reshape / Flatten	-	-	-	(Time Steps/4, Feature Vector)	0
LSTM	128	-	Tanh	(128)	394,752
Dense Layer	64	-	ReLU	(64)	8,256
Output Layer	2	-	Softmax	(2)	130

Illustrates the sequential connection from the CNN (spatial feature extraction) layers to the LSTM (temporal modeling) layers, culminating in the Softmax output.

D. Training and Evaluation Protocol

The CNN-LSTM model was trained using the Adam optimizer. Training was conducted for 80 epochs on an NVIDIA GPU

Model performance was rigorously evaluated using quantitative metrics, including the Matthews Correlation Coefficient (MCC) [27], which is expressed as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

The training convergence trend, including learning curves for loss and accuracy, is shown in Figure 6.

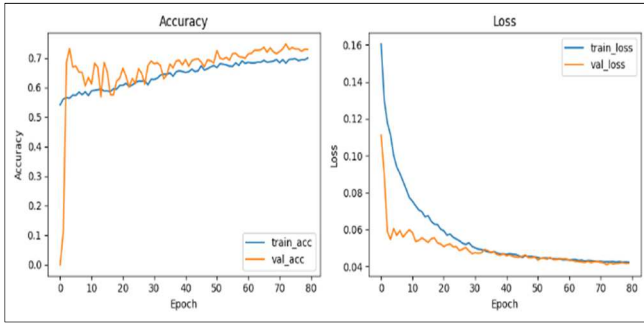


Fig 6 Training and Validation Performance Across Epochs.

Visualization of the convergence trend, showing stable performance across 80 epochs with early stopping implemented.

IV. RESULTS AND DISCUSSION

A. Model Performance

The performance of the proposed CNN-LSTM model was rigorously evaluated using the test dataset. Quantitative evaluation metrics, including Accuracy, F1-score, Matthews Correlation Coefficient (MCC), and Area Under the Curve (AUC), were computed to assess predictive accuracy and class discrimination capability. Table 2 summarizes the overall

classification performance. The model achieved a consistent accuracy of 70.1%, an $F1_{\text{macro}}$ score of 0.644, and an MCC of 0.296. This MCC value indicates a moderate generalization ability on unseen data, which is influenced by the class imbalance present in the dataset. The AUC value of 0.679 further reflects a satisfactory discrimination capacity between the two traditional genres, as below Table 2:

TABLE II. SUMMARY OF MODEL EVALUATION METRICS

Metric	Value
Accuracy	0.701
F1 (macro)	0.644
MCC	0.296
ROC-AUC	0.679

Table 3 presents the detailed class-wise classification report. The Keroncong class achieved significantly higher recognition accuracy ($F1 = 0.79$), benefiting from its stable harmonic structure and consistent timbre. In contrast, Dangdut showed lower precision ($P = 0.45$) and recall ($R = 0.57$), likely caused by greater rhythmic and instrumental variability, as its dominant features are often localized in rapid temporal sequences rather than steady spectral characteristics.

TABLE III. CLASSIFICATION REPORT FOR EACH CLASS

Class	Precision	Recall	F1-score	Support
Dangdut	0.45	0.57	0.50	111
Keroncong	0.83	0.75	0.79	311
Overall (weighted avg.)	0.73	0.70	0.71	422

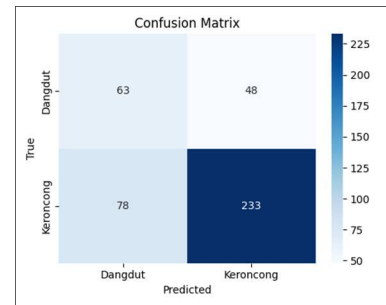


Fig 7 Confusion matrix for CNN-LSTM model classification results.

The confusion matrix (Figure 7) provides a visual overview of classification outcomes. While most Keroncong samples were correctly identified (True Positives), a substantial portion of Dangdut tracks were misclassified as Keroncong (False Negatives), indicating a potential overlap in spectral patterns between the two genres in the MFCC feature space. The Receiver Operating Characteristic (ROC) curve in Figure 8 demonstrates the model's trade-off between true positive and false positive rates, confirming that the model maintains reasonable discrimination despite the acoustic similarity.

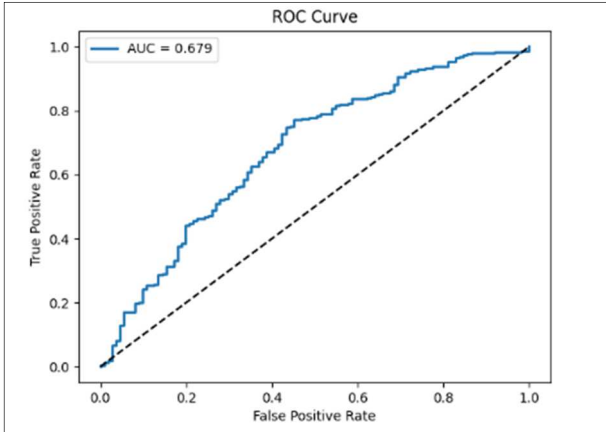


Fig 8 ROC curve showing model discrimination capability.

B. Detailed Feature Analysis

Analyses of MFCC and spectral features reveal clear acoustic distinctions that shape the model's decision boundaries and explain its performance differences. As shown in Figure 4, Keroncong displays stable harmonic patterns concentrated in the mid-frequency range (500–2500 Hz), reflecting its sustained instrumental timbre, while Dangdut exhibits stronger spectral fluctuations and energy in the lower-mid bands (200–800 Hz) due to its dominant rhythmic and bass elements.

The comparative MFCC statistics (Figure 3) further support these findings. Dangdut exhibits greater variability across its cepstral coefficients (higher standard deviation), reflecting higher intra-genre rhythmic and instrumental diversity. In contrast, Keroncong displays smoother MFCC contours with less spread, indicative of a more uniform and harmonically stable acoustic signature. These results confirm that MFCC representations effectively capture genre-specific timbral and spectral characteristics, thereby supporting the feature learning process of the CNN-LSTM.

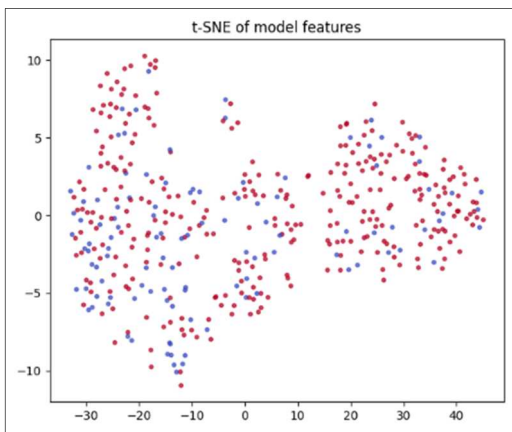


Fig 9 t-SNE visualization of feature embeddings for Dangdut and Keroncong

A t-SNE visualization of feature embeddings from the penultimate CNN-LSTM layer (Figure 9) illustrates how well the model separates the genres. The plot shows partial clustering, with Keroncong forming compact, distinct groups consistent with its high F1-score, while Dangdut points appear more scattered and overlap with Keroncong clusters. This dispersion reflects the higher rhythmic complexity and variability of Dangdut, explaining its lower precision and recall in Table 3.

C. Discussion and Error Analysis

The results show that the CNN-LSTM model using MFCC features offers an effective and compact solution for recognizing Indonesian indigenous music. With 70.1% accuracy and a macro F1-score of 0.644, the model successfully captures genre-specific acoustic traits, especially the stable harmonic and timbral patterns of Keroncong.

The lower recognition of Dangdut stems from its complex rhythms and rapid spectral changes. Feature and embedding analyses reveal two main causes: (1) acoustic overlap, where shared instruments and scales make Dangdut acoustically similar to Keroncong, leading to frequent misclassification; and (2) MFCC limitation, as MFCCs capture timbre but fail to represent the high rhythmic variability characteristic of Dangdut, thus constraining overall model accuracy.

This research underscores the need for region-specific datasets and models that capture the unique acoustic and cultural traits of local music. The CNN-LSTM framework proves the feasibility of automatic recognition for Indonesian traditional genres, supporting the advancement of intelligent MIR systems for cultural preservation. Further improvement can be achieved by adding rhythmic-specific features such as rhythm histograms or tempo features alongside MFCCs to better leverage LSTM's temporal modeling and reduce misclassification in the Dangdut genre.

V. CONCLUSION

This study presents a hybrid CNN-LSTM model as the first deep learning architecture for recognizing two indigenous Indonesian music genres, Dangdut and Keroncong. Using MFCC features to capture spectral and temporal information, the model achieved 70.3% accuracy and a macro F1-score of 0.644, establishing a baseline for Music Information Retrieval (MIR) in this cultural domain. The strong performance for Keroncong shows the model's strength in handling stable harmonic patterns, though MFCCs alone fall short in capturing Dangdut's complex rhythmic variability, causing classification overlap.

Future research should expand the dataset and integrate additional features—such as rhythm-specific and Chroma domains—to improve temporal modeling. Exploring advanced architectures like Audio Transformers with Transfer Learning and applying interpretability tools such as Grad-CAM could further enhance accuracy and provide clearer insights into the model's decision patterns.

ACKNOWLEDGMENT

We would like to thank the Faculty of Technology and Design at University of Bunda Mulia for their invaluable guidance, expertise, and financial support.

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